Appendix A Exercises

State warp size, sector size, launch dimensions, and resource assumptions.

GA-Q01 Grid mapping

Process one million elements with 256 threads/block in a one-dimensional grid.

1. Find the number of blocks.
2. Find useful and out-of-range threads in the final block.
3. Give block/thread/global indices for the first and last elements.
4. With 12 KiB shared memory/block and 48 KiB/SM, all other resources unconstrained, find the shared-memory residency limit.

GA-Q02 Divergence

In a 32-thread warp, five threads take one path and 27 take its mutually exclusive else path. Both paths have equal instruction length.

1. Count masked passes.
2. Compute useful lane-slot utilization.
3. If the else body is absent, find utilization of the if body's instructions.
4. Why can predication fail to improve a long divergent body?

GA-Q03 Coalescing

Thirty-two threads each read a four-byte word using 32-byte sectors.

1. Count sectors for sector-aligned consecutive words.
2. Count sectors and useful-byte fraction with eight-word stride.
3. Count sectors when consecutive access starts four bytes after a sector boundary.
4. Why do few sectors not guarantee saturated DRAM bandwidth?

GA-Q04 Shared banks

Use 32 four-byte banks and word-index-modulo-32 mapping.

1. Analyze `A[base+lane]` across 32 lanes.
2. Analyze `A[base+32*lane]`.
3. Analyze all lanes reading the exact same `A[k]`.
4. Propose a stride-32 transpose layout repair and its remaining costs.

Appendix A Exercises (continued)

GA-Q05 Resource-limited occupancy

An abstract SM supports 768 threads, 24 warps, eight blocks, 32768 registers, and 48 KiB shared. Kernel A uses 256 threads/block, 32 registers/thread, and 16 KiB shared/block. Warp 32; ignore allocation rounding.

1. Find resident-block limits from threads, blocks, registers, and shared memory.
2. Find active threads and occupancy.
3. With 128 threads/block and still 32 registers/thread, compare (a)16 KiB shared/block and (b)64 bytes shared/thread, or 8 KiB/block. Find blocks, warps, occupancy, and limiting resources.
4. Why can 100% occupancy lose to 50%?

GA-Q06 CSR SpMV

```
Ap = [0, 2, 3, 5]
Aj = [0, 2, 1, 0, 2]
Av = [3, -1, 4, 2, 5]
x  = [2, 10, 3]
```

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

1. Give each row's nonzeros.
2. Compute y.
3. Explain lane use and row imbalance with one warp/row.
4. When can staging x in shared or read-only cache help, and when can it lose?

GA-Q07 GPU Roofline

Compute ceiling is 250 GFLOP/s and DRAM bandwidth 320 decimal GB/s.

1. Find ridge intensity.
2. A has .3 FLOP/byte. Find its memory and overall ceilings.
3. B has intensity four. Find its overall bound.
4. A purported A measurement performs 1.2×10^9 FLOP in five ms. Compute the rate and test consistency with A's ceiling. Identify traffic, counting, or phase inputs that must be checked; costs that only slow execution cannot explain exceeding the stated bound.
5. In a separate phase, $F = 10^9$ FLOPs, $P_C = 1000$ GFLOP/s, $Q_D = 0.25 \times 10^9$ bytes, $B_D = 100$ GB/s, $Q_S = 4 \times 10^9$ bytes, and $B_S = 1000$ GB/s. Find both intensities, individual and combined roofs, and the time lower bound.

Appendix A Exercises (continued)

GA-Q08 Unequal divergence

A 32-thread warp has eight lanes on a two-instruction path A and 24 on a six-instruction path B, both nonempty, ignoring branch costs.

1. Find useful lane-instructions, issued slots, and weighted utilization.
2. What does a 50% shortcut omit? If B has no active lanes, how changes the denominator?
3. Why is utilization not CPU-relative kernel speedup? Give three missing costs.

GA-Q09 Sectors and banks

Use 32 lanes, four-byte distinct words, 32-byte sectors, aligned base, no cache. Shared has 32 banks, index modulo 32.

1. For byte addresses 8 i, i 0..31, count sectors/useful fraction.
2. Repeat for 28+4i and stride 128 bytes.
3. For shared word indices 34 i, find banks and words/bank; repeat 35 i. Explain using gcd (stride, 32).
4. Why can padding reduce whole-kernel performance despite fewer conflicts?

GA-Q10 Non-power-of-two scan

Apply Blelloch exclusive scan to [2, 0, 1, 3, 4], padded to eight. All threads obey barriers; integers do not overflow.

1. Give every up/root-zero/down state, first five outputs, and total.
2. Derive addition work/tree span. Why output only five values?
3. Prove down-sweep using the prefix-before-interval invariant.
4. Why is concatenating local block scans not a global scan?

GA-Q11 Stable radix with duplicates

The input [(5, a),(1, b),(5, c),(0, d),(1, e),(7, f),(2, g),(6, h)] has three-bit unsigned keys. Use stable zero-prefix partition from the least bit.

1. For bit 0, give zero flags, prefixes, total, and destinations.
2. Give all three identity-preserving output orders.
3. Prove destination bijection and explain why an unstable later pass can destroy low-bit ordering.

Appendix A Exercises (continued)

GA-Q12 Tiling, conservation, and overall gains

1. Bodies at 0 and 2 have masses 2 and 3, G=1, no softening, self omitted. Find acceleration, force, and total force. Must accelerations sum to zero?
2. Naive reads N sources per target before skipping self plus one my load, no cache. For N=8, B=4, find source/total loads and bytes at 16 bytes/body.
3. For N=10, B=4, how does the final block load tiles and participate? Why is early return unsafe?
4. One input/output pair costs 16.777216 ms, each kernel 2 ms, and each launch 0.02 ms. Keep data for K iterations; CPU takes 25 ms each. Find the least integer K for at least 5 fold end-to-end speedup, no overlap.

References and attribution

David A. Patterson, John L. Hennessy. Computer Organization and Design: The Hardware/Software Interface, Revised Fourth Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study